

AUTOMATIC BOTTLE FILLING USING PLC

Mentor: Jayshree Ramakrishnan

Member name:

Ruchika Pandey: 18

Nimish Pimparkar: 22

Krishna Tikoo: 29



Introduction

Manual bottle filling can lead to inconsistent filling, spillage, and human error. Our project provides an automated solution using PLC-based filling control and OpenCV-based quality inspection to ensure consistent filling and detect spillage.

- An automated system designed to detect, fill, and inspect bottles with minimal human intervention.
 - A PLC-based control system manages the conveyor, bottle detection, and water-filling process.
 - A proximity sensor detects the bottle and triggers the filling sequence for a predefined time.
 - A camera and OpenCV-based vision system checks the filled water level and detects possible spillage.
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OpenCV-Based Quality Inspection

A camera captures the filled bottle at the inspection platform.
OpenCV processes the image to detect the water level and spillage.
The water level is compared with the required filling range.
Based on the results, the bottle is classified as PASS or FAIL.

Flow:

Camera → Image Processing → Level & Spillage Detection → PASS/FAIL

Current Status

- PLC-Based Control Architecture:

PLC control for the conveyor and water pump has been finalized, including the required inputs, outputs, and sequence.

- Components Identified:

The main components such as PLC, sensor, conveyor, pump, relay, and power supply have been identified.

- Conveyor System Planned:

The conveyor is planned to move and position bottles automatically using PLC control.



Current Status

- Bottle Detection Sensor:

Sensor selected to detect bottles at the filling position and provide input to the PLC.

- Relay Module:

Relay module selected for safe control of the conveyor motor and water pump.

- Basic PLC Ladder Logic:

Basic logic developed for start/stop, bottle detection, conveyor, and pump control.

- PLC & Hardware Integration:

PLC programming, hardware integration, and component testing are currently in progress.





Milestone Target & Deadline

Finalize Project Design & Components = April 2026 – Research Started July 2026

Procure Required Components = August 2026

Build Conveyor & Filling Mechanism = September 2026

Develop PLC Program= September 2026

Integrate Sensor, Pump & Conveyor= October 2026

Develop OpenCV Bottle Detection= October 2026

Complete Testing & Debugging= November 2026

Final Demonstration & Documentation= November 2026





THANK YOU

-RUCHIKA PANDEY